GCSE Chemistry
Exam Walkthrough Pack

Step-by-step worked exam-style questions with full explanations

Original JDScience exam-style questions — not copied from past papers

Introduction

This Chemistry Exam Walkthrough Pack is designed to help you understand how to approach GCSE and IGCSE exam-style questions. Every question in this booklet is an original JDScience item written to match common specification skills — it is not copied from AQA, Edexcel, OCR or other past papers.

How to use this pack: read each question, cover the walkthrough, and try answering in exam conditions first. Then read what the question is asking, follow the step-by-step thinking, compare your answer with the model answer, and study the mark breakdown to see exactly how marks are awarded.

Command words tell you what type of answer is required. Section 1 explains the most common GCSE science command words. Always underline the command word before you start writing.

Showing working matters: in calculations, examiners award method marks even when the final answer is wrong. In longer answers, a clear logical sequence helps you secure every available mark.

Examiners award marks for correct science linked to the question. Keywords from the specification matter, but a correct idea in your own words can still score. Extended questions often use level-based marking: plan before you write, use paragraphs, and link ideas.

To move from Grade 4/5 to Grade 7/8/9: use precise terminology, explain mechanisms (not just names), include data or examples where asked, and check units and significant figures in calculations. For six-mark questions, a short plan and linked paragraphs beat a long unordered list.

Section 1: Command word guide

State

What you must do: Give a short, factual answer — often one word, number or phrase. No explanation is required unless the question also asks you to explain.

Common mistake: Writing a long paragraph when a single fact is enough, or giving an explanation when only a fact was asked for.

Model sentence starter: The [quantity/name] is …

Give

What you must do: Provide the information requested. Similar to state, but may need a short phrase or named example.

Common mistake: Listing unrelated facts that do not answer the question.

Model sentence starter: One example is … / The value is …

Describe

What you must do: Say what happens or what something is like. Focus on observations, patterns or features. You do not need to say why.

Common mistake: Explaining causes when the command word is only describe.

Model sentence starter: First … then … / The pattern shows …

Explain

What you must do: Give reasons why something happens. Link cause and effect using science ideas and key terms.

Common mistake: Describing what happens without saying why, or using everyday language instead of science.

Model sentence starter: This happens because … which means …

Calculate

What you must do: Work out a numerical answer. Show your working, include units, and give the final value to the required precision.

Common mistake: Missing units, no working, or using the wrong equation.

Model sentence starter: Use … = … / Substitute: …

Compare

What you must do: Identify similarities and/or differences between two or more things. Refer to both sides.

Common mistake: Only describing one item, or listing features without comparing.

Model sentence starter: Both … however … / X has … whereas Y …

Evaluate

What you must do: Weigh up evidence or options, consider strengths and weaknesses, and reach a supported judgement.

Common mistake: Only listing advantages with no conclusion or balanced comment.

Model sentence starter: An advantage is … A limitation is … Overall …

Suggest

What you must do: Apply your knowledge to a new situation. Your answer should be sensible and scientifically plausible.

Common mistake: Repeating textbook facts that do not fit the context given.

Model sentence starter: A possible reason is … / This could be because …

Justify

What you must do: Give evidence or reasons that support a choice, conclusion or statement already made.

Common mistake: Restating the claim without giving supporting science.

Model sentence starter: This is supported by … because …

Determine

What you must do: Find out a value or conclusion using data, a graph or a calculation. Show how you reached your answer.

Common mistake: Reading a graph incorrectly or not showing how the value was obtained.

Model sentence starter: From the graph … / Using the data …

Analyse

What you must do: Break information into parts, interpret patterns or trends, and explain what the data shows.

Common mistake: Describing data without interpreting it, or ignoring anomalies.

Model sentence starter: The data shows that … / As X increases, Y …

Section 2: Atomic structure and the periodic table

QUESTION 1

State the relative charges of a proton, neutron and electron.

QUESTION TYPE

Topic: Atomic structure and the periodic table | **Command word:** state | **Marks:** 1 | **Skill tested:** Atomic structure

WHAT THE QUESTION IS ASKING

Recall the three subatomic charges tested at GCSE.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Proton: +1; neutron: 0; electron: –1.

MARK BREAKDOWN

Mark 1: Proton: +1; neutron: 0; electron: –1.

COMMON MISTAKES

- Writing neutrons as –1
- Mixing up proton and electron charges

EXAMINER TIP

List particles in the order proton, neutron, electron so you do not miss one.

QUESTION 2

Give the number of protons, neutrons and electrons in an atom of carbon-12.

QUESTION TYPE

Topic: Atomic structure and the periodic table | **Command word:** give | **Marks:** 1 | **Skill tested:** Atomic structure

WHAT THE QUESTION IS ASKING

Use the mass number and atomic number for carbon.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

6 protons, 6 neutrons, 6 electrons.

MARK BREAKDOWN

Mark 1: 6 protons, 6 neutrons, 6 electrons.

COMMON MISTAKES

- Using 12 neutrons
- Forgetting electrons equal protons in a neutral atom

EXAMINER TIP

Neutrons = mass number – atomic number.

QUESTION 3

State what is meant by the term isotope.

QUESTION TYPE

Topic: Atomic structure and the periodic table | **Command word:** state | **Marks:** 1 | **Skill tested:** Atomic structure

WHAT THE QUESTION IS ASKING

Define isotopes in terms of atomic structure.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Isotopes are atoms of the same element with the same number of protons but different numbers of neutrons.

MARK BREAKDOWN

Mark 1: Isotopes are atoms of the same element with the same number of protons but different numbers of neutrons.

COMMON MISTAKES

- Saying isotopes have different numbers of protons
- Only mentioning mass number without protons

EXAMINER TIP

Always include 'same element' and 'different neutrons'.

QUESTION 4

Describe the plum pudding model of the atom.

QUESTION TYPE

Topic: Atomic structure and the periodic table | **Command word:** describe | **Marks:** 2 | **Skill tested:** Models of the atom

WHAT THE QUESTION IS ASKING

Outline the key features of Thomson's model.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The atom is a ball of positive charge with negative electrons embedded in it, like plums in a pudding.

MARK BREAKDOWN

Mark 1: Ball of positive charge spread through the atom

Mark 2: Negative electrons embedded in the positive charge

COMMON MISTAKES

- Calling it the nuclear model
- Forgetting that the positive charge was spread out

EXAMINER TIP

Link the name 'plum pudding' to embedded electrons.

QUESTION 5

Explain why atoms are electrically neutral.

QUESTION TYPE

Topic: Atomic structure and the periodic table | **Command word:** explain | **Marks:** 2 | **Skill tested:** Atomic structure

WHAT THE QUESTION IS ASKING

Relate proton and electron numbers to overall charge.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

An atom has equal numbers of protons (+1) and electrons (–1), so the charges cancel.

MARK BREAKDOWN

Mark 1: Equal numbers of protons and neutrons

Mark 2: No charged particles present

COMMON MISTAKES

- Only stating 'no charge' without numbers
- Mentioning neutrons as charged

EXAMINER TIP

Use both particle names and their charges.

QUESTION 6

Give two differences between Group 1 and Group 7 elements.

QUESTION TYPE

Topic: Atomic structure and the periodic table | **Command word:** give | **Marks:** 2 | **Skill tested:** Periodic table

WHAT THE QUESTION IS ASKING

Compare reactivity and electron arrangement across groups.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Group 1 metals are more reactive down the group and form 1+ ions; Group 7 non-metals are less reactive down the group and form 1– ions.

MARK BREAKDOWN

Mark 1: Group 1 reactivity decreases down the group

Mark 2: Group 7 elements are metals

COMMON MISTAKES

- Comparing periods instead of groups
- Ignoring ion formation

EXAMINER TIP

Mention reactivity trend and ion charge for both groups.

QUESTION 7

Explain how the results of the alpha particle scattering experiment led to the nuclear model.

QUESTION TYPE

Topic: Atomic structure and the periodic table | **Command word:** explain | **Marks:** 3 | **Skill tested:** Models of the atom

WHAT THE QUESTION IS ASKING

Link observations (most pass through, some deflect, few rebound) to structure.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Most alpha particles passed straight through, showing mostly empty space. Some were deflected, showing a positive nucleus. A few rebounded, showing a small, dense, positive centre.

MARK BREAKDOWN

Mark 1: Most particles passed through → mostly empty space

Mark 2: Deflection/rebound → small dense positive nucleus

Mark 3: Nuclear model replaces plum pudding

COMMON MISTAKES

- Saying electrons caused deflection
- Claiming the atom is solid

EXAMINER TIP

Use the three observations in order: pass through, deflect, rebound.

QUESTION 8

Describe the electron arrangement in sodium and explain why it forms a 1+ ion.

QUESTION TYPE

Topic: Atomic structure and the periodic table | **Command word:** describe | **Marks:** 3 | **Skill tested:** Ions

WHAT THE QUESTION IS ASKING

State configuration 2,8,1 and link to losing one electron.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Sodium has electron arrangement 2,8,1. It loses one outer electron to achieve a full outer shell, forming Na■ with a 1+ charge.